5-2. Regression illustration using R

Enter a small example data set:

```
> x=c(3.1,3.4,3.0,3.2,3.9,2.8,2.2)
> y=c(3.5,3.2,3.0,2.9,4.0,2.5,2.3)
```

Take a look at the data:

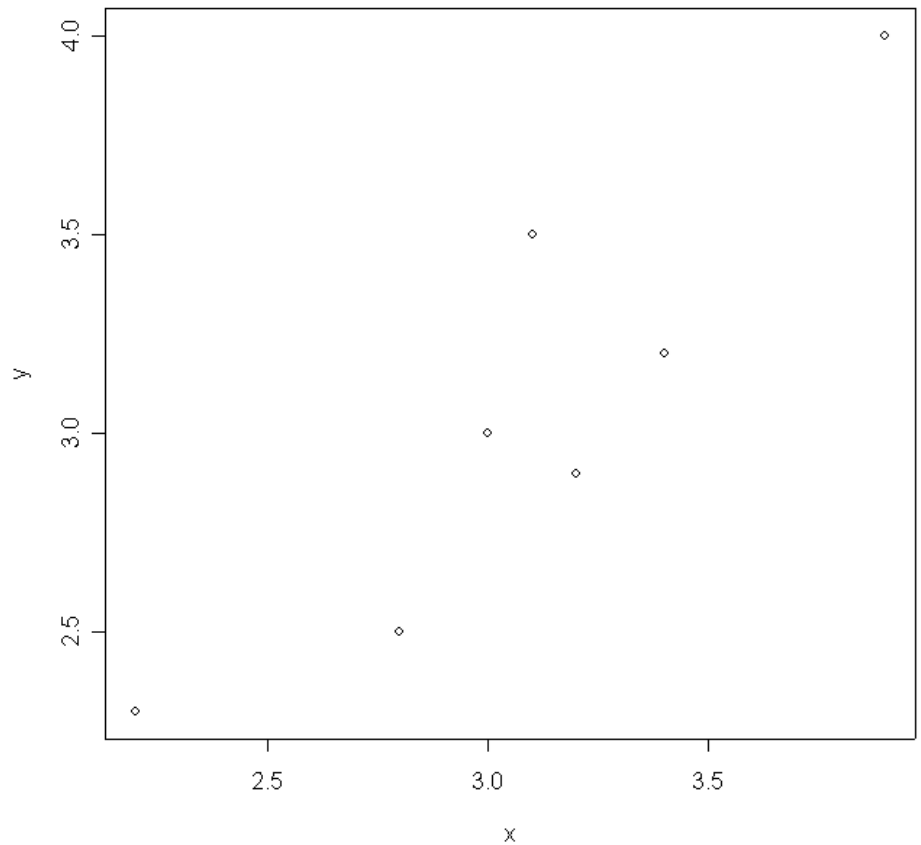
```
> x
[1] 3.1 3.4 3.0 3.2 3.9 2.8 2.2
> y
[1] 3.5 3.2 3.0 2.9 4.0 2.5 2.3
```

Now take a look at a scatterplot:

```
> plot(x, y)
```

or

```
> plot(y ~ x)
```



Now use the `lm` command to fit the regression model, and examine the result.

```
> example1 = lm(y ~ x)
> example1

Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
  -0.004506     0.992201
```

So the fitted model is

$$\hat{y} = -0.004506 + 0.992201x$$

Can get a lot more information using `summary(example1)`:

```
> summary(example1)

Call:
lm(formula = y ~ x)

Residuals:
      1       2       3       4       5       6       7
 0.4287 -0.1690  0.0279 -0.2705  0.1349 -0.2737  0.1217

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.004506   0.683082  -0.007  0.99499
x            0.992201   0.218681   4.537  0.00618 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2808 on 5 degrees of freedom
Multiple R-squared:  0.8046,Adjusted R-squared:  0.7655
F-statistic: 20.59 on 1 and 5 DF,  p-value: 0.006185
```

Examine fitted values:

```
> yhat = fitted.values(example1)
> yhat
```

```
       1        2        3        4        5        6        7
3.071317 3.368977 2.972097 3.170537 3.865078 2.773657 2.178336
```

Alternatively:

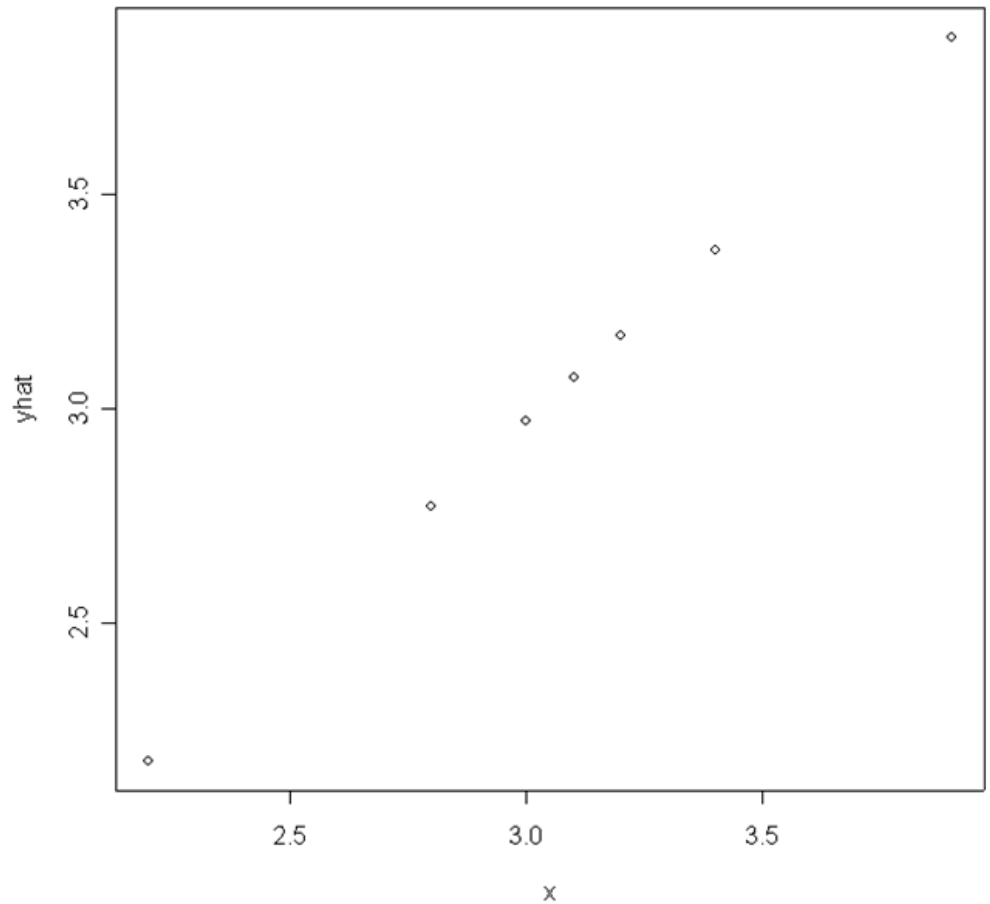
```
> ahat = -0.004506
> bhat = 0.992201
> yhat = ahat + bhat*x
> yhat
```

```
[1] 3.071317 3.368977 2.972097 3.170537 3.865078 2.773657 2.178336
```

Plot fitted values, `yhat`, against explanatory variable values, x.

This in fact gives points along the fitted line.

```
> plot(yhat ~ x)
```



Next examine residuals:

```
> epsilon = residuals(example1)
> epsilon
```

```
          1           2           3           4           5
 0.42868284 -0.16897747  0.02790295 -0.27053726  0.13492201
          6           7
-0.27365685  0.12166378
```

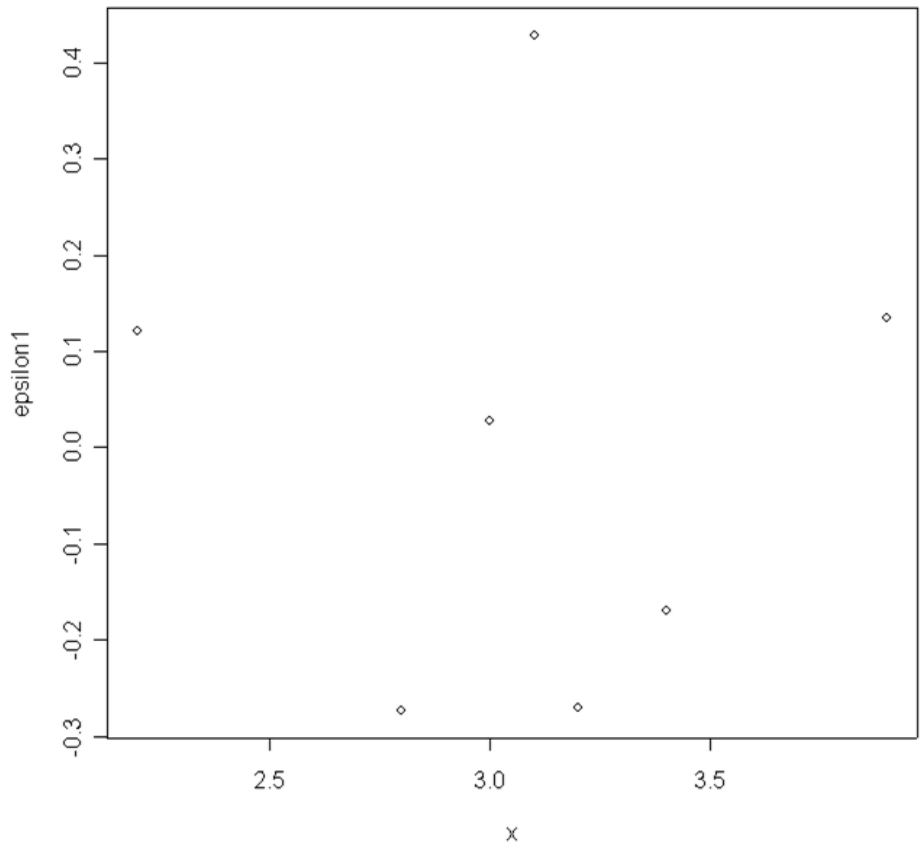
Alternatively:

```
> epsilon = y - yhat
> epsilon
```

```
[1]  0.42868284 -0.16897747  0.02790295 -0.27053726  0.13492201
[6] -0.27365685  0.12166378
```

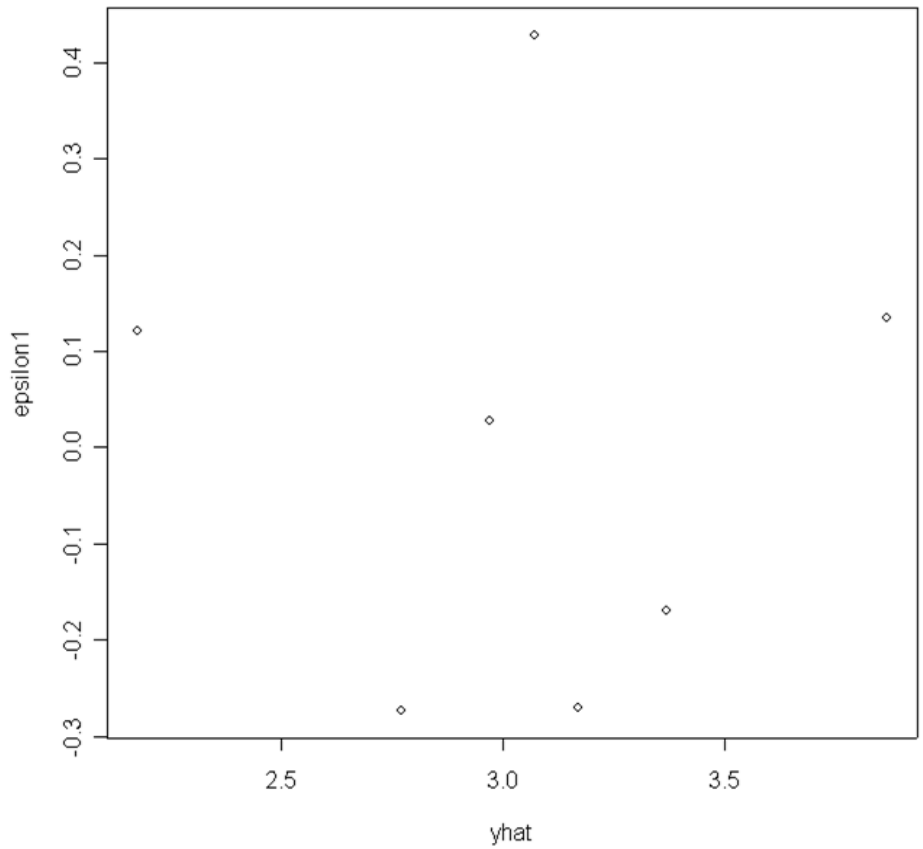
Plot residuals against the explanatory variable x:

```
> plot(epsilon ~ x)
```



Plot residuals against the fitted values `yhat`:

```
> plot(epsilon ~ yhat)
```

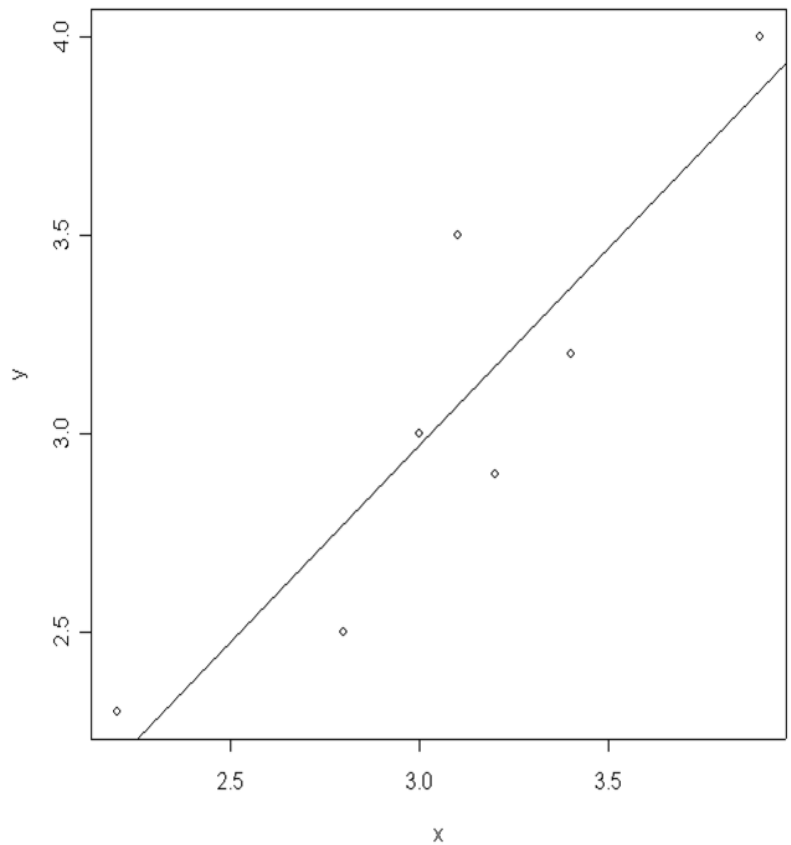


Both graphs show the same thing — the residuals follow a random pattern.

Note: Since the fitted line is approximately $\hat{y} = x$, both graphs are extremely similar in this case.

Plot y against x with fitted regression line:

```
> plot(y ~ x)
> abline(example1)
```



Prediction in R

Predict y at $x = 3.1$, and obtain 95% Confidence Interval (CI) for *mean response*:

```
> predict(example1, data.frame(x=3.1), se.fit=T,
interval="confidence")


$fit
       fit      lwr      upr
1 3.071317 2.798397 3.344237

$se.fit
[1] 0.1061706

$df
[1] 5

$residual.scale
[1] 0.2807794
```

Take a look at the R help for this command using ?predict.lm

Prediction in R

Now obtain 95% Prediction Interval at $x = 3.1$, sometimes called CI for *individual response*:

```
> predict(example1, data.frame(x=3.1), interval="prediction")
```

```
       fit      lwr     upr
1 3.071317 2.299675 3.84296
```

Notice that the point estimate $\hat{y} = 3.0713$ is the same as before, but that the Prediction Interval $(2.300, 3.843)$ for an individual response is wider than the Confidence Interval $(2.798, 3.344)$ for the mean response.

Sum of squares

'Analysis of Variance'

```
> summary.aov(example1)

            Df Sum Sq Mean Sq F value  Pr(>F)
x            1 1.6230  1.6230   20.59 0.00618 **
Residuals    5 0.3942  0.0788
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$SS_{REG} = 1.6230$

$SS_{RES} = 0.3942$